

Statistical and Computational Models

Lecture 7: Binary outcomes

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Roadmap

Modelling a probability

Logit and probit

Fitting

Binary outcomes

Very often we encounter outcomes of the form:

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$$Y_i \in \{0, 1\}$$

Examples in political science: **many**

- Turnout
- Roll-call votes
- Vote choice (more later)
- Interstate conflict
- Alliances
- Regime type (1 = democracy)
- etc. etc.

Binary outcomes

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Target: the **conditional** probability $\mathbb{E}[Y_i | \mathbf{x}_i] = \mathbb{P}(Y_i = 1 | \mathbf{x}_i)$.

The linear probability model

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One option, g^{-1} the identity, is the **linear probability model** (LPM):

$$\mathbb{P}(Y_i = 1 \mid \mathbf{x}_i) = \mathbf{x}_i^\top \beta, \quad \text{or equivalently} \quad Y_i \sim \text{Bernoulli}(\pi_i), \quad \pi_i = \mathbf{x}_i^\top \beta$$

Read: the probability that $Y_i = 1$ comes from a weighted sum of the covariates $\mathbf{x}_i$.

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$\hat{\beta}_{\text{oil}} = -0.22$: at the same `lgdp`, oil states are 22 percentage points less likely to be democracies.

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$$\text{Var}[\varepsilon_i | \mathbf{x}_i] = \text{Var}[Y_i | \mathbf{x}_i] = \mathbf{x}_i^\top \beta (1 - \mathbf{x}_i^\top \beta)$$

Read: the variance $\pi_i(1 - \pi_i)$ moves with $\mathbf{x}_i$.

Guidance for the LPM

Despite these downsides, the simplicity of the LPM means it is very popular.

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1. For **average effects** with one binary, ignorable x , the model is **saturated**, so β is exactly the ATE.
2. Generally, LPM slopes are often quite similar to alternative models; it is a good idea to **check and report alternative models** even if you use an LPM.
3. Always use **heteroskedasticity-robust** standard errors, `{sandwich}::vcovHC()` or `{estimatr}::lm_robust()`.

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The **CDF**: defined for every $\mathbf{z} \in \mathbb{R}$, between 0 and 1, never decreasing.

Aside: the CDF

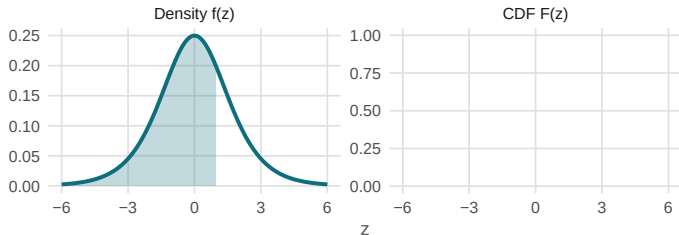
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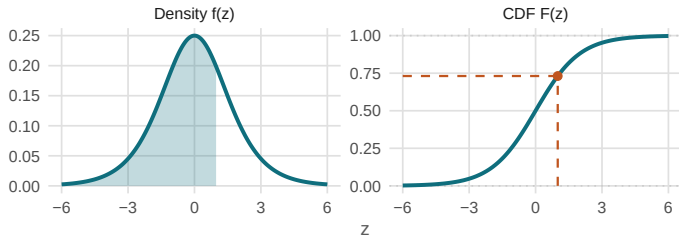


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Example: the standard **logistic**. Shaded area left of $z = 1$: $F(1) = 0.73$.

Keeping $\hat{Y}$ in $[0,1]$

Solution: make g^{-1} a **CDF**, F :

$$\mathbb{P}(Y_i = 1 \mid \mathbf{x}_i) = g^{-1}(\mathbf{x}_i^\top \beta) = F(\mathbf{x}_i^\top \beta), \quad \text{logit}^{-1}(z) = \frac{e^z}{1 + e^z}$$

Logit: logistic CDF, $F = \text{logit}^{-1}$ (a.k.a. **inverse logit**).

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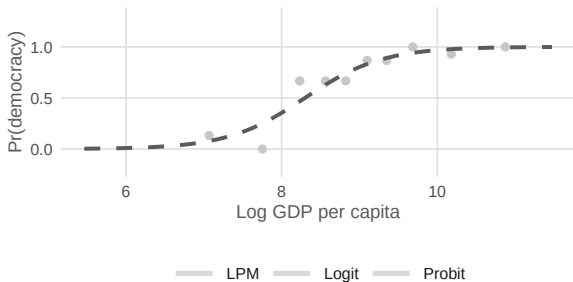
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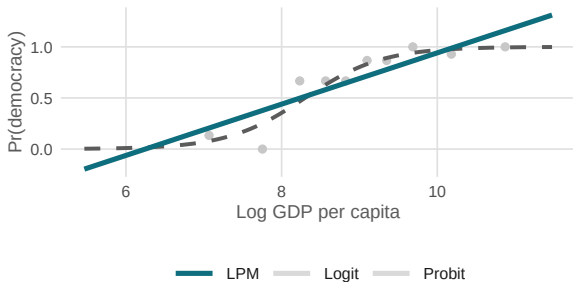
Note: F is **not** the CDF of Y_i , which is Bernoulli (using a CDF is just a convenient way to ensure $0 \leq \pi_i \leq 1$).

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Grey dots: binned means at $\text{oil} = 0$. Dashed: the true DGP.

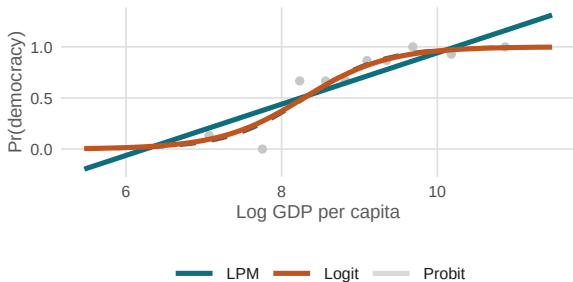
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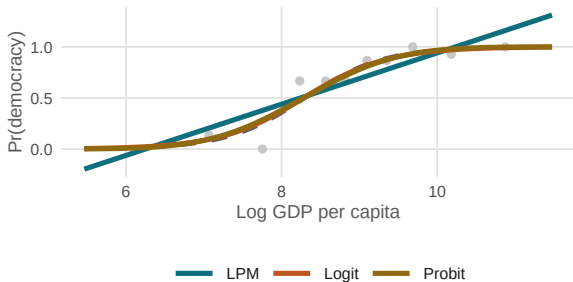


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Logit, and probit: little practical difference, but note that $\hat{\beta}$ are completely different between the models.

A generalized linear model

Recall: $Y_i \sim \text{Bernoulli}(\pi_i)$. And we want π_i to be a function of $\mathbf{x}_i$:

1. A distribution from the exponential family (Bernoulli, normal, ...).
2. A linear predictor: $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$.
3. A link: $g(\mathbb{E}[Y_i | \mathbf{x}_i]) = \eta_i$.

g is the **link**. We have been working with its inverse, $\mathbb{E}[Y_i | \mathbf{x}_i] = g^{-1}(\eta_i)$.

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This is a **generalized linear model** (GLM; Rodríguez, App. B).

From distribution to link

Write the Bernoulli in **exponential-family** form, $\exp\{y\theta - b(\theta) + c(y)\}$:

$$\pi_i^{y_i}(1 - \pi_i)^{1-y_i} = \exp \left\{ y_i \underbrace{\log \frac{\pi_i}{1 - \pi_i}}_{\theta_i} + \log(1 - \pi_i) \right\}$$

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The **canonical link** sets $\theta_i = \eta_i$, so g is read off the distribution:

Y_i	Mean	θ	Canonical g	Model
Normal	μ	μ	identity	OLS
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Canonical is a convenience (e.g. probit uses $\mathbf{g} = \Phi^{-1}$).

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Assume that Y_i is generated by a distribution that depends on covariates $\mathbf{x}_i$ and some parameters (that we care about).

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$$L(\beta; \mathbf{Y}, \mathbf{X}) = \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i}, \quad \pi_i = \text{logit}^{-1}(\mathbf{x}_i^\top \beta)$$
$$\ell(\beta) = \sum_{i=1}^n \left\{ y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i) \right\}$$

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At $\hat{\beta}$, $\ell = -86.72$. At our no-covariate $\hat{\pi}$, it was -138.74 .

The likelihood, with covariates

Recall the **score**, the derivative of ℓ in β (after tedious math):

$$\mathbf{s}(\beta) = \sum_{i=1}^n (y_i - \pi_i) \mathbf{x}_i$$

Set $\mathbf{s}(\beta)$ to zero and solve for $\hat{\beta}$. With only an intercept, $\sum_i (y_i - \pi) = 0$ gives lecture 6's $\hat{\pi} = r/n$.

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But π_i is **nonlinear** in β : no closed form. Instead, we use lecture 6's Newton–Raphson (`glm()`: Fisher scoring).

Fitting it

Our point estimates and standard errors:

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logit <- glm(y ~ oil + lgdp, data = countries, family = binomial,  
            control = glm.control(epsilon = 1e-10))  
round(summary(logit)$coefficients[, 1:2], 3)
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	Estimate	Std. Error
(Intercept)	-15.329	2.343
oil	-1.560	0.429
lgdp	1.851	0.273

`family = binomial` with a 0/1 y : the Bernoulli (one trial), with the logit link by default.

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No estimate is more than one se from the truth.

References

- Rodríguez, G. *Generalized Linear Models*, Princeton lecture notes – **3.1–3.3, 3.7**, 3.4–3.6, App. B. grodriguez.github.io/glms.